

Name: Anthony Claudet

Date: _____

Class: _____

MyDesign Portfolio

Elements G-I (combined in one doc)

Element G: Construction of a Testable Prototype

Final Prototype Iteration Explanation (G1)

Prototype Construction of Design #1 (make as many copies, updating the design number, as needed)

Date: 5/6/25

Details about all steps of construction (**add step numbers as needed**), including the tools and manufacturing methods used:

	Details	Tools and Manufacturing Methods used
Step 1	In a new Matlab live script, create a flexible structure to store each device's name, power (kW), essential status, and new field duration (hours).	Matlab struct arrays
Step 2	Classify peak vs. off-peak automatically by using the system	Matlab datetime and hour() functions

Name: Anthony Claudet

Date: _____

Class: _____

	clock rather than manual input.	
Step 3	Collect data for each device and validate data types (e.g., power must be numeric, is or is not essential).	input() function in Matlab
Step 4	Loop through each device in the array and check the time and if it's non essential. Provide recommendations accordingly.	For loops, booleans, conditionals
Step 5	For every device recommended to be delayed, calculate potential kWh saved by multiplying power by duration.	Basic multiplication
Step 6	Display personalized recommendations for each device. Print total estimated kWh saved if delay recommendations are followed.	Formatted output and printing
Step 7	Test scalability and see if the program can handle up to 10+ devices. Look for	Loop testing, struct arrays

Name: Anthony Claudet

Date: _____

Class: _____

	bugs or incorrect outputs.	
--	----------------------------	--

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

- Classifies time automatically by using Matlab's system clock
- Uses kWh instead of kW for saved energy units
- Added basic data checking and better savings estimates
- Tested for 10+ devices instead of just 5



STOP HERE—Create the initial prototype and test it. After you have iterated on the initial design, you will complete the second part of this document. You may iterate more than once.

Final Prototype Construction of Design #2

Date: 5/6/25

Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration: n/a

Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

	Details	Tools and Manufacturing Methods used
Step 1	Create a structure to store each device's data- name, power, essential, and status	Matlab struct or table

Name: Anthony Claudet

Date: _____

Class: _____

	(delay, turn off, turn on).	
Step 2	Create a dashboard layout with sections for device list, time display, and color-coded recommendations.	Matlab UI tools
Step 3	Automate time detection and classify peak hours	Matlab datetime and hour() functions
Step 4	Use input fields to allow device data entry	GUI text boxes, dropdown menus, dialogs
Step 5	Evaluate devices and update visual indicators such as box colors	Matlab UI tools
Step 6	Test if the GUI handles 10+ devices smoothly	Scrollable panels
Step 7		

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

- Uses 10+ devices instead of 5
- Automates time and classifies hours
- Includes a summary box and device status display